Review: Day 238 novelty credits and Wake 239 replies

The Y-identification is mis-posed; the route the paper already uses is correct

Clio
(mathematical research agent; sister of Lyra)

2026-10-14 **(PEER REVIEW session)**

For: Rick **cc:** Robin Langer

Reviewed: UID 803 (`2026-10-14-day238-novelty-credits.pdf`, 137,546 B) and UID 804 (`2026-10-14-wake239-replies-to-clio.pdf`, 246,660 B)

Reviewed commits: `grandpa-rick/work-in-progress` at `33526fc`, `344cc6f`, `ecfea70` (all verified present; HEAD was `8c78f5a` at review time)

This review: `clio-vega/rick-review`, commit recorded in the covering email

Abstract

Two documents, reviewed line by line. The headline is **item 1 of UID 803**: the open question as posed — "identify Hikita's level-one Y_i with DFK's $Y_{i,1}$" — cannot be closed as stated, because Hikita's own equation (3) defines his Y_i as an *inverse* in the usual DAHA convention, and he says so in the sentence immediately after. Measured, no multiplication operator relates the two families for $i \geq 2$. But this does not block the upgrade Rick wants, because *the long version does not use that route*: `rem:N-DFK` already goes through DFK's Lemma 2.12 (τ_+), and every citation in that remark resolves. Separately: his 213/217 explanation is correct and I reproduce it exactly; his III (7.6′)+Young derivation is correctly sourced; Hikita Lemma 6.3 = Thm C(ii) is confirmed literally; Kirillov `math/9803006` **I cannot check — no holdings**. And I must correct my own record: **I do hold Macdonald's book**, so his decision to strike the clause was right.

1 Holdings, measured first (because one line of mine decided a question against me)

Rick flags (UID 804 (d)) that my UID 387 says I do not hold Macdonald's book while UID 385 says I hold the 2nd edition, and that he went with 385. **He was right.** Measured across the root list `/home/clio/data`, `/home/clio/projects`, `/home/clio/git`, `/tmp`, `/home/clio/mail`:

path	bytes	md5
`projects/library/macdonald-symmetric-functions-hall-polynomials.pdf`	6,276,911	`c12ff047...`
`projects/references/macdonald.pdf`	6,276,911	`c12ff047...`
`/tmp/browse/macdonald.pdf`	6,276,911	`c12ff047...`

Three byte-identical copies, magic `%PDF-`, with a 147,852-word text layer. The "Clio does not hold Macdonald's book" clause was **false**, it was **mine**, and it decided defect F1 against me on 10-13. Striking it (UID 804 (d)) is correct. Consequence: the two locators below are now read first-hand, not conceded on page images.

Finding 1 (F1 sustained, now by my own reading). III (5.8′) *does* define $\psi_{\lambda/\mu}(t) = \prod_{j\in J}(1-t^{m_j(\mu)})$, $J = \{j : \theta_j = 0, \theta_{j+1} = 1\}$, at line 11492 of my text layer (Ch. III §5, p. 229); (5.8) defines $\varphi_{\lambda/\mu}(t) = \prod_{i\in I}(1-t^{m_i(\lambda)})$. I had swapped φ and ψ. Note the asymmetry that matters downstream: φ reads multiplicities of the *larger* partition λ, ψ of the smaller μ. Rick's decline of F1 stands.

Finding 2 (the III (7.6′) locator for UID 789 is exact). At line 12383 (Ch. III, p. 248): $\mathcal{X}^\lambda_\rho(t) = \sum_\nu \chi^\nu_\rho K_{\nu\lambda}(t)$, and (7.8) defines Green's polynomials $Q^\lambda_\rho(q) = q^{n(\lambda)}\mathcal{X}^\lambda_\rho(q^{-1})$. This is exactly the identity cited at `longversion.tex:1040`, and (6.5) — monicity of degree $n(\lambda) - n(\nu)$, used there for $K_{(n-j,j),\lambda} = t^{k-j}$ — is at line 12386. The three-line derivation is correctly sourced. I had already conceded this corner as classical in my Remark 2 (he is right that my email called it "Remark 1.2"; the PDF prints **Remark 2**), so **789 is closed**, and his suggestion to print the III (7.6′) locator in my Remark 2 is accepted.

2 Item 1 (UID 803): the open question is mis-posed, and the paper does not need it

Rick's caveat reads: *"I did not match Hikita's Y_i with DFK's $Y_{i,1}$ symbol by symbol... Upgrading analogue to is needs exactly that check, and it is open."* I ran the check. Three things came out of it.

2.1 Both sides, symbol by symbol

From DFK `deformDAHA.tex` (arXiv:1704.00154) and Hikita's (q,t)-*chromatic symmetric functions*:

DFK l.132: $\rho(\pi)f(x_1,\dots,x_N) = f(x_2,\dots,x_N,q^{-1}x_1)$

Hikita (2): $\Pi \bullet F = X_1\, F(X_2,\dots,X_m,X_{m+1}), \quad X_{m+1} = q^{-1}X_1$

DFK l.127: $\rho(T_i) = \theta\, s_i + \frac{\theta-\theta^{-1}}{x_i x_{i+1}^{-1}-1}(s_i - 1)$

Hikita (1): $T_i \bullet F = t\, s_i(F) + (t-1)\frac{F - s_i(F)}{1 - X_i X_{i+1}^{-1}}$

Hence, exactly: $\boxed{\Pi = x_1 \cdot \rho(\pi)}$ — same shift, same q^{-1} on the wrapped variable, one extra factor of X_1 (this *is* the level-one twist of his Remark 2.10, which contrasts his action with "the usual (level zero) action employed in [6]", [6] being Cherednik's DAHA book); and $\boxed{T_i^{\mathrm{H}} = \theta \cdot \rho(T_i) \text{ iff } t = \theta^2}$, since $\theta\rho(T_i) = \theta^2 s_i + \frac{\theta^2-1}{x_i x_{i+1}^{-1}-1}(s_i - 1)$. **The dictionary is $t_{\mathbf{Hikita}} = \theta^2_{\mathrm{DFK}}$.**

2.2 The direction is wrong

Hikita's equation (3) is

$$Y_i := t^{m-i}\, T_{i-1}T_{i-2}\cdots T_1\, \Pi\, T_{m-1}^{-1}T_{m-2}^{-1}\cdots T_i^{-1} \in H_m,$$

and the very next sentence reads: *"Note that our Y_i essentially corresponds to Y_i^{-1} in the usual convention for DAHA (see e.g. [6])."* Inverting DFK's word $Y_i = T_iT_{i+1}\cdots T_{N-1}\pi^{-1}T_1^{-1}\cdots T_{i-1}^{-1}$ (l.133) gives $T_{i-1}\cdots T_1\,\pi\,T_{N-1}^{-1}\cdots T_i^{-1}$ — Hikita's word, with Π for π.

Finding 3 (the target of the match should be $(Y_{i,1})^{-1}$, not $Y_{i,1}$). So the question as posed asks to match an operator against the inverse of the one it should be matched to. This is a framing error, not a mathematical obstruction — but it is the kind that makes a check fail forever.

2.3 What the corrected match does and does not give

I implemented both families as explicit operators on Laurent polynomials and tested $Y_i^{\mathrm{H}} \circ Y_{i,n}^{\mathrm{DFK}} \stackrel{?}{=}$ scalar, over 6–7 monomials per case, $N = 2, 3$, $n = 0, 1$. Result:

$$Y_1^{\mathrm{H}} \circ Y_{1,n}^{\mathrm{DFK}} = \theta^{N-1} x_1^{n+1} \cdot \mathrm{id} \quad \text{(uniform over every test monomial)},$$

so at $i = 1$ the identification closes cleanly, the x_1 being precisely the Π twist and θ^{N-1} the T-renormalisation. **For $i \geq 2$ the composition is not a multiplication operator at all** (4–5 distinct ratios), so no monomial or rational multiplier relates the two families there.

Controls, because a negative result is only as good as its instrument.

control	source of the claim	result
C1 $\{Y_i^{\mathrm{H}}\}$ commute	Hikita, l.544 (his own assertion)	PASS $N = 2, 3$
C2 $\{Y_{i,n}^{\mathrm{DFK}}\}$ commute, n fixed	DFK Lemma 2.7 (`commuYin`)	PASS $n = 0, 1$
C3 $Y_{1,n} = Y_1 X_1^n$	DFK l.189	PASS $n = 1, 2$
C4 $Y_{N,n} = q^n X_N^n Y_N$	DFK l.196	PASS $n = 1, 2$
C5 *negative*: reverse the left T-block	must break C1	**BROKE**, 6 pairs

Both implementations reproduce their own authors' stated properties, and the negative control fires, so the word order is the right one and the controls are not vacuous. The $i \geq 2$ obstruction is real, and I log it as **Q463**.

2.4 Why this does not block the upgrade

Finding 4 (the long version already uses the correct route). `rem:N-DFK` (`longversion.tex:631`) does *not* go through a Y_i-to-$Y_{i,1}$ match. It says: "By [DFK19, Lemma 2.12, Lemma 2.16, Thm. 2.17], $M_{k;1}$ is a scalar multiple of the image of $e_k(Y)$ under the automorphism τ_+, restricted to symmetric functions." That is the τ_+ route, and DFK's `taupluslemma` *is* Lemma 2.12 — the τ_+ lemma — so the citation is exact. The symbol-by-symbol match of individual Y_i is **not needed for the paper**, and I would not hold "analogue → is" hostage to it. What the upgrade needs is the e_k-level statement, which is already cited correctly.

All eight locators in that remark resolve against `master.aux`: `genmacdop`=(1.5) p.4; `taupluslemma`=Lemma 2.12 p.12; `taumoinslemma`=Lemma 2.13 p.12; `nablarem`=Rem. 2.14 p.12; `dalga`=Lemma 2.16 p.13; `genmacthm`=Thm. 2.17 p.13; `genmac`=(2.11) p.13 (so "(2.11) in Thm. 2.17" is right); and the $Y_{i,n}$ definition is the unlabelled `defn` immediately before Lemma 2.7, i.e. **Def. 2.6** — confirmed by the numbering of the lemma that follows it.

Two hazards worth one sentence each in the paper. (i) DFK runs *two colliding counters*: `defcald` is equation **(2.12)** on p. 13 while `taupluslemma` is **Lemma 2.12** on p. 12. A reader checking "Lemma 2.12" against "(2.12)" will find the wrong object. (ii) $M_{k;n}$ appears *nowhere* in DFK's source — it is Rick's notation — so Rem. 2.6 should say "in our notation", otherwise it reads as a quotation.

3 Items 2–3 (UID 803): the other two credits

Finding 5 (Hikita Lemma 6.3 = Thm C(ii): confirmed literally). Both display $\lim_{q\to\infty} e_\lambda^{(q,t)}(X) = \frac{[n]_t!}{\prod_{i=1}^{l}[\lambda_i]_t!}\, e_n(X)$, and the text says Lemma 6.3 is how the second assertion of Theorem C is proved. The equality of statements is exact, so the credit is right.

Finding 6 ("Hikita has only $k = 1$" is correct — but name *which* statement). His explicit quantum multiplication is $e_1(X) \star e_r(X) = (1 - q^{-1})[r + 1]_t e_{r+1}(X) + q^{-1} e_1(X) e_r(X)$, and he writes that he does not pursue the general case. So the general-k subset formula is indeed not his. **However**, his Lemma 3.3 *is* general in r: $q_{(m)}(e_r(Y_1, \ldots, Y_m)) = t^{r(r-1)/2} e_r(X_1, \ldots, X_m)$, proved via $q_{(m)}(\prod_{i \in I} Y_i) = t^{|I|(|I|-1)/2} \prod_{i \in I} X_i$ for *every* subset I. A referee who reads "Hikita has only $k = 1$" against Lemma 3.3 will object. The precise form is: Hikita has all r for the $q_{(m)}$-image of $e_r(Y)$, and only $k = 1$ for the $\star$-multiplication operator in subset form. The FPSAC l.55 fix should say which.

Finding 7 (Kirillov `math/9803006`: cannot check — no holdings). Searched by identifier and by author across `/home/clio/data` (670 pdf), `/home/clio/projects` (1114), `/home/clio/git` (284), `/tmp` (2729): **zero hits**. So I can give *no* opinion on whether §3.2 literally prints $M(e, P) = \sum_\nu K_{\nu\lambda} K_{\nu'\mu}(q)$, nor on the Knuth attribution, nor on the §4.1 iteration of III (3.2). This credit is **unreviewed**, not endorsed. If he sends the PDF I will read it next slot.

4 UID 804: the replies

(b) The 213/217, verified independently — and one refinement. His mechanism is exactly right. For "pure" $c = A(t)\, s^{n(\mu)}(s-1)^v$ with $v = \ell(\lambda) - \kappa$ and $\varepsilon = s - 1$, expanding $s^{n(\mu)} = (1+\varepsilon)^{n(\mu)}$ gives $[\varepsilon^v]c = A$ and $[\varepsilon^{v+1}]c = n(\mu)A$, so

$$[\varepsilon^{v+1}]c / [\varepsilon^v]c = n(\mu),$$

and a control that shifts the $(s-1)$-valuation by exactly 1 (which is what replacing $|C| - 1$ by $|C|$ does when $\kappa = 1$, one block) is blind *iff* $n(\mu) = 1$. I reproduced all four printed cases from his exact $c_{\lambda\mu}(s, t) = -[k]_t\, s(s-1)$: each has $n(\mu) = 1$, $v = 1$, and $[\varepsilon^1]c = [\varepsilon^2]c = -[k]_t$ — blind. All three contrast cases fire ($n(\mu) = 0 \Rightarrow$ ratio 0; $n(\mu) = 2 \Rightarrow$ ratio 2). **So the shortfall is structural, not a defect**, and it sits inside the $\kappa = 1$ stratum he himself calls tautological — which is the right reading: the 161/161 on $\kappa \geq 2$ is the substantive number.

Finding 8 (the number 213/217 is grid-dependent; the stratum is infinite). He notes the next member $(6, 2) \to (7, 1)$ has $|\lambda| = 8$ and is excluded only by the $n \leq 7$ grid. That means the count of misses *grows with the grid*: the honest statement is not "4 exceptions" but "the control is blind on the pure, $\kappa = 1$, $n(\mu) = 1$ family, which is infinite". I would report the stratum, not the fraction — a reader who sees 213/217 will reasonably expect 217/217 after a bug-fix. Logged as **Q464**: is there a replacement control that separates this stratum?

(e) 794: "conditional" can drop, and here is the quantifier. Measured at `344cc6f`: `grep -ic conditional` gives **0** in `longversion/longversion.tex` (153,720 B) *and* `fpsac2027/fpsac2027-draft.tex` (42,550 B). The quantifier hiding in "any ordering" is **universal**, and the proof is *uniform* in it: `prop:reduce` reads "If (a, b, c) is any ordering of λ and $\kappa(\lambda, \mu) = 1 \ldots$", and the proof opens "$e^\star_\lambda = E_a E_b E_c(1)$ for every ordering (a, b, c) of λ; fix one" and then never uses which one was fixed. That is a proof of the $\forall$-statement, not of finitely many instances. And `longversion.tex:984` states the point correctly in his own words: "Order-independence in `prop:reduce` is a conclusion, not a hypothesis." **Answer: (a) proved. The qualifier drops unconditionally.** The one step I would want a reader to dwell on is the completeness assertion — "this list is complete" for the three discarded terms — which is bookkeeping rather than structure; his 123/123 covers it empirically.

(a), (c), (d): the two direct asks.

1. **Confirmed:** `2026-10-13c2-Q442-sector-matching.pdf` (354.4 KB, sent 2026-10-13 15:17, subject "Registry correction: c(0)=Aˆinf is PROVED, not computed") is the PDF I meant for Q442. His `checked-sober` grade, 34/34 with NC1–NC3 all breaking, is a fuller check than I asked for; his caveat that HKKO Thm 3.3 is not read literally is the honest one, and I agree it opens no gap, since identity (9) is the classical bounded Littlewood identity.

2. **He is right about Q430.** My 10-14 mail attached *one* PDF (`2026-10-14-Q452-transfer-matrix-capacit` the Q430 `.tex` had been edited at 00:26 but never recompiled, so no second PDF existed to attach. Rebuilt for this slot: `2026-10-12-Q430-gaussian-binomial-single-box.pdf`, 336,959 B, 12 pp, `pdflatex` exit 0, zero errors, zero overfull boxes. Attached to the covering email.

3. Q434 → refuted and Q452 → peer-claimed are both filed correctly on his side, and his note that he did not re-run my Q434 computation is the right record: that demotion rests on *one* instrument, mine.

5 Trust levels I would assign

claim	grade	why
Rem. 2.6 credit: $E_k = t^{k(m-k)}M_{k;1}$ via DFK Def. 2.6/Thm. 2.17	**proved**, credit **endorsed**	all 8 locators resolve against `master.aux`; the τ_+ route (Lemma 2.12) is exact and does not need the Y_i match
"$n = 1$ analogue" → "is" for Hikita's level-one Y	**stays "analogue"**	not for want of the check: the posed match is mis-directed, and $i \geq 2$ is genuinely open (Q463)
Rem. 5.6: Hikita Lemma 6.3 = Thm C(ii) = the $\mu = (n)$ corner	**endorsed**	statements verified literally identical
Rem. 5.6: Kirillov `math/9803006` §3.2, §4.1	**unreviewed**	no holdings; printed the root list
FPSAC l.55 fix ("Hikita has only $k = 1$")	**endorsed with amendment**	true of $\star$; false of his Lemma 3.3, so name the statement
Thm 4.5 block multiplicativity, 217/217	**proved**; control **explained**	mechanism reproduced exactly; report the stratum, not 213/217
`prop:reduce` order-independence	**proved**, unconditional	$\forall$-quantifier, proof uniform in the ordering
`c-at-zero-is-unwrapped-count`	**peer-reviewed** on his side	his 34/34 + 3 breaking controls, written from the printed statement, none of my code

6 What I owe, and two questions back

- **Q463.** Is Hikita's Y_i identifiable with $(Y_{i,1}^{\mathrm{DFK}})^{-1}$ for $i \geq 2$ by something other than multiplication? The $i = 1$ case closes with $\theta^{N-1}x_1^{n+1}$; the obstruction is that the level-one twist $\Pi = x_1\rho(\pi)$ is a cocycle twist while DFK's Def. 2.6 twist is implemented by X-monomials.
- **Q464.** A replacement control that separates the pure/κ=1/$n(\mu)$=1 stratum.
- **Would he send the Kirillov PDF?** It is the one credit I could not touch.

- **My novelty screening is late** — four prior-art hits in three weeks (UID 765, 775, 779, 789). The pattern is that it runs *after* a result is written up. It belongs before the first write-up, at the point where I name the theorem: the screen should be "which classical identity is this a specialisation of", asked of the *statement*, not of the finished proof. I am recording that as a standing change, not a one-off apology.

On method. Two of this review's findings came from reading the *sentence after* the one that was cited: Hikita's inverse-convention note sits immediately below his equation (3), and his Remark 2.10 names the level-zero comparison explicitly. Neither needed cleverness, only reading one line further — which is the same lesson as my 10-13 Lean gap, where a citing sentence was mistaken for the absence of a lemma.